

Backend Project Synopsis Document

University Name: Atria University

Department: Digital transformation

Course: Backend Development – Node.js, Express, MongoDB

Project Title: MinorMap+ : Smart Minor Eligibility & Recommendation Engine

Submitted by: Team 500 OK

Team Lead: Ibrahim

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1. Project Overview

MinorMap+ is a backend-driven recommendation system designed to assist university students in identifying which academic minor fields best align with their strengths, interests, and aptitude. The platform allows students to register, log in, select a minor track, and complete a structured quiz. Their responses are evaluated using a weighted scoring algorithm that calculates suitability across multiple minors.

The backend is built using **Node.js**, **Express.js**, and **MongoDB** with **JWT-based authentication**. Students receive categorized results as **Eligible**, **Potential**, or **Not Recommended**, along with the top recommended minors. Admin users are provided with capabilities to manage quiz questions, scoring weight configurations, and view analytics trends.

The core purpose of MinorMap+ is to enable data-driven academic decision-making by offering personalized recommendations, reducing confusion among students, and supporting advisors in guiding career-aligned minor selection.

2. Motivation and Problem Statement

University students often struggle to choose appropriate minors due to limited awareness of how their skill sets align with available specialization tracks. Many students depend on peer suggestions or guesswork without a structured evaluation method. This leads to **misaligned academic pathways**, reduced motivation, and inefficient utilization of educational opportunities.

MinorMap+ addresses this gap by providing a **systematic and intelligent evaluation model**. By analyzing quiz responses through a weighted eligibility algorithm, the system generates reliable recommendations based on measurable traits rather than subjective assumptions.

The project is relevant to modern academic environments that aim to provide personalized support and career guidance. Through backend-controlled logic and transparent scoring, the system benefits **students, faculty advisors, and career counselors** with insights that are easy to interpret and apply. MinorMap+ serves as a stepping stone toward building **AI-assisted academic guidance systems**.

3. Objectives

- Develop secure backend APIs for user authentication and minor recommendation workflows.
- Implement a weighted scoring algorithm to compute eligibility for various minors.
- Provide admin functionality to manage quiz questions and minor information.
- Ensure secure user access using JWT authentication and hashed credentials.
- Validate and test routes thoroughly using Postman, ensuring reliability and data integrity.

4. System Feasibility

Technical Feasibility:

The system uses widely adopted backend technologies (Node.js, Express.js, MongoDB, JWT), which are stable, well-documented, and supported by the development team's skill set.

Operational Feasibility:

Deployment can be executed on any cloud/VM environment supporting Node.js. The design is modular, scalable, and maintainable.

Economic Feasibility:

The solution leverages open-source tools. No licensing costs are required. Hardware and operational expenses remain minimal.

5. Proposed System Design

System Architecture (Conceptual):

User → Frontend (HTML/CSS/JS) → API Layer (Express.js)
→ Business Logic (Controllers + Scoring Engine)
→ MongoDB (users, questions, minors, recommendations)

Module Description:

- **Authentication Module:** Register and login with password hashing & JWT.
 - **Quiz Module:** Retrieve questions and store responses.
 - **Recommendation Module:** Weighted scoring algorithm outputs eligibility.
 - **Admin Panel Module:** Manage question bank and minor data.
 - **Middleware:** Token verification, logging, validation, and error handling.
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6. Methodology

1. Requirement Gathering
2. System & Database Design
3. API Development (Controllers, Routes, Models)
4. Scoring Algorithm Implementation
5. Testing with Postman Collections
6. Frontend Integration
7. Deployment & Documentation

7. Expected Deliverables

- Fully functional backend server with secure REST APIs.
- MongoDB database schemas and relationships.
- JWT-based login and authorization system.
- Six working frontend pages (Register, Login, Select Minor, Quiz, Results).
- Postman API test collection.
- Final documentation and submission report.

8. Novelty and Impact

MinorMap+ introduces **automated academic guidance** tailored to individual competency profiles. Unlike traditional manual counseling, this system ensures objectivity and fairness by using algorithmic evaluation rather than assumptions. The transparency of the scoring rules and interpretable recommendation results support student confidence and advisor decision-making. This promotes informed minor selection and improved academic alignment.

9. Future Scope

- Integration with Learning Management Systems for automatic performance analysis.
- AI-based adaptive quiz difficulty and personalized questioning logic.
- Dashboard analytics for faculty trend insights.
- Mobile app integration for accessibility and engagement.

10. References

- Node.js Documentation: <https://nodejs.org>
- Express.js Documentation: <https://expressjs.com>
- MongoDB Manual: <https://www.mongodb.com/docs>
- JWT: <https://jwt.io>